Prepared for referee review

RESPONSE TO THE R97 ADVISORY REFEREE REPORT ON NEURAL BELLMAN OPERATORS

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This revision addresses the report deposited on 14 October 2026 at commit `9723c6c`. It develops the original Neural Bellman Operators paper rather than replacing its topic. New results concern action-aligned qualification, nonuniform and randomized information, and an exact economic realization of acquired continuation geometry. The original adverse investment evidence and all inherited substantive sections are preserved. This is a response to the repository's advisory review, not a representation that Econometrica has issued an editorial decision or accepted the revision.

1. THE REVISED CONTRIBUTION

The report correctly distinguishes the validity of a certificate from the usefulness of the representation to which it is attached. It also identifies a decisive empirical fact: residual-span improvement in all 36 matched cells accompanies an increase in first-proposal rejections from 644 to 757. We retain that evidence and respond by constructing the missing action-level link, not by renaming a residual improvement as a policy improvement.

The original subject remains Neural Bellman Operators: own-future construction, feasible original owners, complete implemented policies, recursive original-law risk and the resource account. Three developments strengthen that subject. Theorem 7 connects an outward supporting calculation to the original all-action gap. Theorem 9 and Proposition 10 extend the information analysis with explicit law and randomization quantifiers. Theorem 11 establishes exact policy-loss separation for the original oblique continuation in a specified economic decision problem. All proofs are supplied, and the exact acquisition and economic identities have two separately implemented rational calculations.

2. RESPONSES TO BLOCKING CONCERNS AND REQUESTED REVISIONS

B1; M1 and M4: Neural specificity and the original subject. We retain the NBO construction rather than recast the paper as an unrelated method-neutral certificate paper. The constructed neural continuation and its feasible owner are genuine objects of the original problem. At the same time, no theorem can distinguish a ReLU from a conventional implementation of the identical hinge. Section 9 therefore reports the neural representation, native hinge, two-triangle representation and a dictionary containing that hinge together. The best additive and quadratic comparators have positive exact policy losses; equal oblique geometry has equal zero loss. This is a decision-theoretic value of the acquired geometry, not an unsupported claim of superiority over every conventional method.

B2; M2: The information theorem was too narrow. Theorem 9 gives lower and upper information bounds under a known density bounded above and away from zero. It reduces exactly to the prior uniform-law constants. Proposition 10 distinguishes pathwise-sound randomized intervals from ordinary randomized point estimates. For the latter, a fixed Rademacher distribution of convex tail perturbations yields an explicit $N^{-5/2}$ worst-case root-mean-square lower bound. The proof is not the invalid procedure of choosing a different adverse input after observing each random tape. We claim neither a matching randomized upper rate nor dimension-independent Bellman complexity. Equation (32) supplies a constructive dated query allocation on a fixed graph, with all additional graph and tensor costs retained. The prior deterministic theorem and proof remain complete in the Development Appendix.

B3; M3 and M6: Approximation separation was not a policy result. Theorem 11 remedies this directly. The benefit is the shifted original oblique ridge and the posted-price law is fixed before the calculation. Conditional price integration gives exactly $(h - v)^2/6$. The optimal additive and quadratic expected regrets are $1/540$ and $1/5400$, with explicit feasible population projections. The oblique representation changes actual purchase decisions and eliminates this loss. The acquisition threshold in (38) prices that identified gain. The construction is theoretical and explicitly

normalized; it is not presented as an empirical calibration of the separate investment economy.

B4 and B10; M3 and M5: Residual-oriented fitting failed its service connection. The smaller residual spans and worse rejections are preserved. Theorem 7 supplies the missing sufficient relationship: an original-objective supporting quadratic and an upper policy value bound the original action gap. The slope mismatch and constrained stationarity residual enter quadratically, while enclosure width is charged separately. The risk-witness lift shows how the same feasible original-law witnesses used in the investment construction generate this support. Theorem 8 then gives an out-of-sample rejection guarantee for empirical rejection selection over an independently frozen menu, or a bounded-work guarantee when complete work is selected directly. The bounded surrogate result is stated against its own surrogate optimum, not falsely against the best rejection rate. No newly proposed selector is attributed to the old frozen investment services. The supplement also supplies an exact counterexample showing strict residual-oscillation improvement with worse action loss.

B5; M4, M7 and M8: Complete work and empirical scope. Quadratic least squares remains the measured investment winner. All first-proposal rejections, repair work, refinements, fallback, acquisition charges and operation counts remain in the retained tables and source records. The exact oblique mechanism adds conventional geometry controls analytically; it is not a new wall-clock tournament or an adaptive-geometry acquisition campaign. This revision does not report an unexecuted higher-dimensional, longer-horizon or larger-action experiment. The original scalar/vector horizon theorem and adverse executed vector evidence remain intact. The new dated precision calculation states its graph-size assumptions and counts additional work. Historical clock ranges remain descriptive machine-specific observations, not population confidence intervals.

B6; M6 and M8: Economic purchase and identification. The investment price-grid exercise remains conditional sensitivity and still has zero certified neural adoption regions over quadratic least squares. Its per-decision policy contrast is multiplied

by the number of decisions in each timed workload, and acquisition intervals are subtracted by interval arithmetic. The new purchase mechanism has an exactly identified policy contrast and a transparent fixed-cost threshold, but no external monetary or hardware-price calibration is manufactured. These two economic exercises are identified separately.

B7; M5: Population inference for the 84 new services. The 84 acquisition configurations remain a diagnostic catalogue. The earlier frozen-actor confidence intervals are not transferred to them. Their deterministic original-optimum qualification continues to provide the stated distribution-free policy-difference support. The prospective selection theorem requires an independent validation sample for a fixed menu; it does not itself constitute such a sample. The existing R98 research protocol, selected actors and source commitments are preserved unchanged, and its fresh test is not consumed or represented as completed in this revision.

B8; M9: Independent acquisition and dependency closure. The new exact mechanism has two separate acquisition implementations. One builds monomial Gram equations and uses rational elimination; the other uses a triangular transformation, rational Newton–Cotes integration and orthogonal Legendre projection. They share the rational arithmetic kernel but not fitting, integration or selection routines. They reproduce the fitted coefficients, original-law costs, domain range and strict purchase boundary. Fifteen deterministic test methods also cover constrained actions, risk atoms, nonuniform-law enclosures, convex bump constants and a deliberately perturbed coefficient. This is algorithmic independence within an AI-assisted revision, not an external human replication. It does not close independent refitting of the historical nonlinear feature search. The complete inherited executable source is retained with the revision, and the new mechanism needs no third-party Python package. The release identifies exactly which commands were executed and which historical campaigns were not rerun.

B9; M10: Active manuscript and historical preservation. The article's mathematical path is reorganized around construction, original-law qualification, decision alignment, information and economic comparison. The complete expectation-occupation,

signed-risk, decision-precision and earlier information/acquisition sections move to the Development Appendix, together with their full proofs. The risk-occupation and horizon tails are split exactly between the active paper and companion; their concatenation reproduces the original source. All original section, code and table bytes are checked against the source copy; the original active entry points and editorial framing are archived verbatim. A machine-readable retention map and a reader-facing content map identify every relocation. This avoids arbitrary deletion. It does not claim that retaining a large development companion by itself satisfies the report's request for a short overall project; the actual lengths and remaining editorial choice are exposed rather than hidden by smaller type or altered margins.

3. MINOR COMMENTS, PROVENANCE AND NEXT REVIEW

The abstract now separates deterministic information exponent from ordinary randomized lower bounds and states the equal-geometry comparison. The law-moment oracle, risk-level dependence, original-action quantifier, fitting information and complete-policy account are stated alongside the respective results. HJB verification, derivative-informed dynamic programming and decision-aware learning are added to the literature comparison. The ridge example is no longer used as an approximation-only proxy for economic loss. Rejection outcomes remain adjacent to residual-span results, and descriptive clocks are not called confidence intervals.

The revision protocol was deposited before executing the new exact mechanism. The current branch descends from the pre-existing R98 research head, while the scientific manuscript baseline is the precise R97 head reviewed in the report. No review branch, historical manuscript, test stream or main branch is overwritten. The source, executable tests, compiled article, Technical Supplement, Development Appendix and this response are supplied for another substantive review. The new mathematical arguments are fully stated and testable; the revision does not substitute compilation or regression tests for the referee's assessment of their correctness, novelty and economic importance.